

# Radio Environment Map as Enabler for Practical Cognitive Radio Networks

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## ABSTRACT

Latest regulations on TV white space communications and emerging trends of spectrum access through geolocation databases relax the regulatory constraints on Cognitive Radios (CRs). Geolocation databases are designed to store information related to incumbents, and CRs are envisioned to consult this database before spectrum access. Spectrum occupancy and related environment information can be constructed using these geolocation databases. In that regard, Radio Environment Map (REM) is a promising tool that provides a practical means for the realization of cognitive radio networks (CRNs). It constructs a comprehensive map of the CRN by utilizing multi-domain information from geolocation databases, characteristics of spectrum use, geographical terrain models, propagation environment, and regulations. REMs contribute to cognition engines by building long-term knowledge via processing spectrum measurements collected from sensors to estimate the state of locations where there is no measurement data. In addition, REM utilizes feedback from the CRs and can apply various learning tools. The vision is to design CRNs such that CRs, though being simple devices without advanced cognitive functionalities, can become cognitive via REMs and operate in an efficient manner. In this paper, an overview of the REM concept is presented in various dimensions ranging from its architecture and stored information to REM construction techniques as well as REM quality metrics.

## INTRODUCTION

Research on cognitive radio (CR) has matured from early works on the core idea of CRs to current multidimensional proposals such as signal processing techniques, machine learning algorithms, and regulations for cognitive radio. However, there still remain many technical challenges to overcome for evolving CRs into practical networks. TV White Space (TVWS), the portion of the UHF and VHF bands emerging after digital switch-over, has been an effective catalyst for the first practical applications of Cognitive Radio Networks (CRNs).

As the first step of opportunistic spectrum

access, a TVWS device must establish the available frequencies at its location just before communications and must ensure non-harming the primary users (PUs). Due to the concerns mainly raised by the broadcasting industry on the proper protection of the incumbent TV services, regulatory bodies have brought strict rules on the white space devices (WSD). The rules demanded by the regulators such as the detection of a primary signal at  $-114$  dBm by FCC, are overly conservative. Such regulations reduce the available white spaces drastically (e.g., by a factor of three [1]). Instead, accessing a centralized database which stores available spectrum based on geographical coordinates or the properties of registered PU transmitters has been accepted more promising for mitigating these challenges while meeting the regulatory obligations. Currently, database-based TVWS operation is being considered by regulators in the US and the UK. In November 2009, FCC opened a call for database administrators, and in 2010 approved ten companies (e.g., Google, Spectrum Bridge, Telcordia, etc.), four of which completed the tests by 2013.

In a database-based spectrum access scheme, CR devices first query the database of the available frequencies according to their geolocation in terms of latitude and longitude, and in return receive the list of unoccupied frequencies before initiating a communication. TVWS communications make use of this approach via TV Databases (TVDB), which are also known as White Space Databases (WSDB). Since spectrum activities at TV bands are quasi-static and PU transmitters are static, TVDB can meet the requirements of TVWS communications. On the other hand, it may fall short of providing sufficient information in a “real” dynamic spectrum access (DSA) network in which PUs are mobile and frequently change their activities. A step further from geolocation database lies the Radio Environment Map (REM). Contrary to the geolocation DB, REM is an advanced knowledge base that stores live multi-domain information on the entities in the network as well as the environment historically. Therefore, REMs can act as an enabler for cognition in radios and considering all these capabilities of REM, it would not be overestimating to claim that REMs are essential components of practical CRNs.

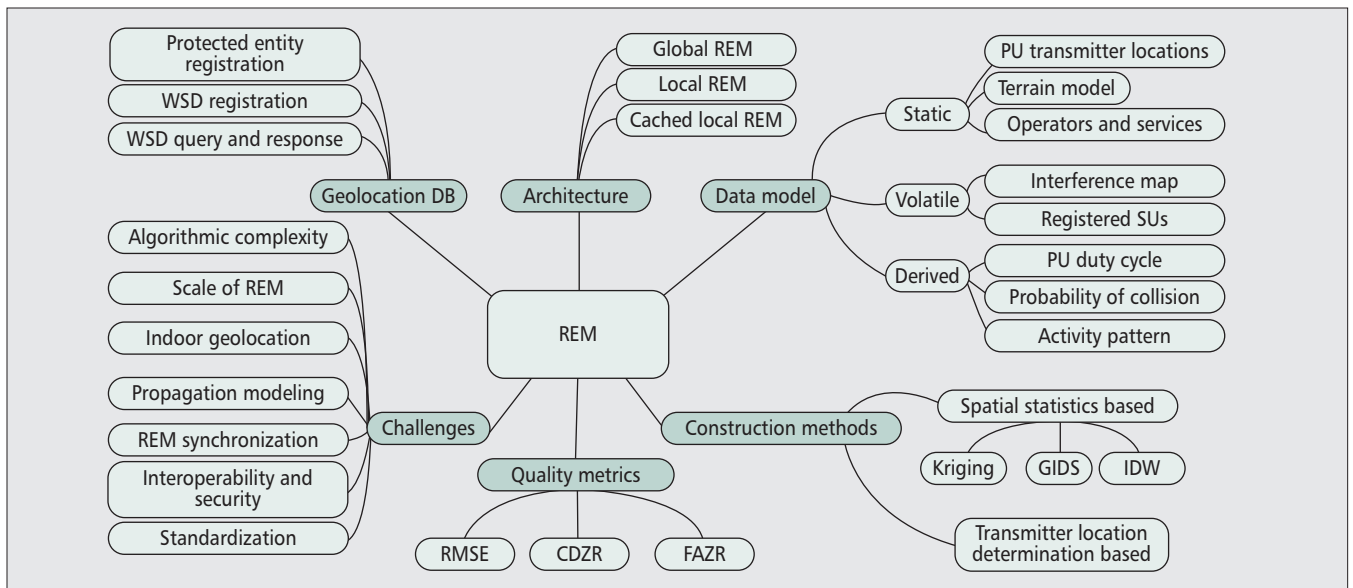


Figure 1. Various aspects of REM.

Essential functionality of a REM is the construction of dynamic interference map for each frequency at each location of interest. A REM collects spectrum measurements from the nodes with spectrum sensing capabilities (i.e. CRs in the network or dedicated spectrum sensors, which are referred to as measurement capable devices, MCD) either periodically or based on REM's decisions for a reliable data fusion. Since it is impractical to have measurements at each location in the CR operation area, REM fuses the available measurements to estimate the interference level at locations with no measurement data. Various methods exist in the literature for REM construction, which can be broadly categorized as *spatial statistics based approaches* and *transmitter location determination based approaches*.

In this article, we elaborate on various aspects of REMs through a holistic view; data stored in REMs, system architecture, REM construction methods in the literature, quality metrics of these methods, and challenges for efficient REM performance are all presented. Figure 1 summarizes the content of this article. We mainly focus on the evaluation of REM construction techniques and refer new metrics for measuring REM quality.

## REM BASED SPECTRUM ACCESS

### GEOLOCATION DATABASE

Geolocation database keeps various information on PU (and possibly CR) transmitters. For instance, in case of TVWS, TVDB stores TV tower locations, antenna heights, user types (PU or CR), device transmitter power, technology, operation channel(s), and duration of use [3]. Basically, a geolocation-database consists of three services: *data repository*, *registrant*, and *query service*. Data repository is the database where permanent service and user information are stored while registrant service provides the registration of both the protected entities (e.g.

TV towers in a TVWS network) and the CRs. Query service provides both the CR and the public interface. The CR query service receives the CR related information and returns the available channel list while the public interface provides a web based system for public access.

Figure 2 presents an example TVDB architecture and provided services, that reflects the TVDB proposal in [2]. On December 22, 2011, FCC approved Spectrum Bridge's database proposal, and starting from January 2012, all incumbents including wireless microphones are mandated to register to this database in order to be protected from secondary access. Currently, several Internet web sites<sup>1</sup> provide the coverage area of TV stations at a specific location, visualize them on a map, and also present the list of available channels. REM concept carries this process a step further and continuously monitors the environment while merging the multi-domain information, which is more appropriate for dynamic frequency bands.

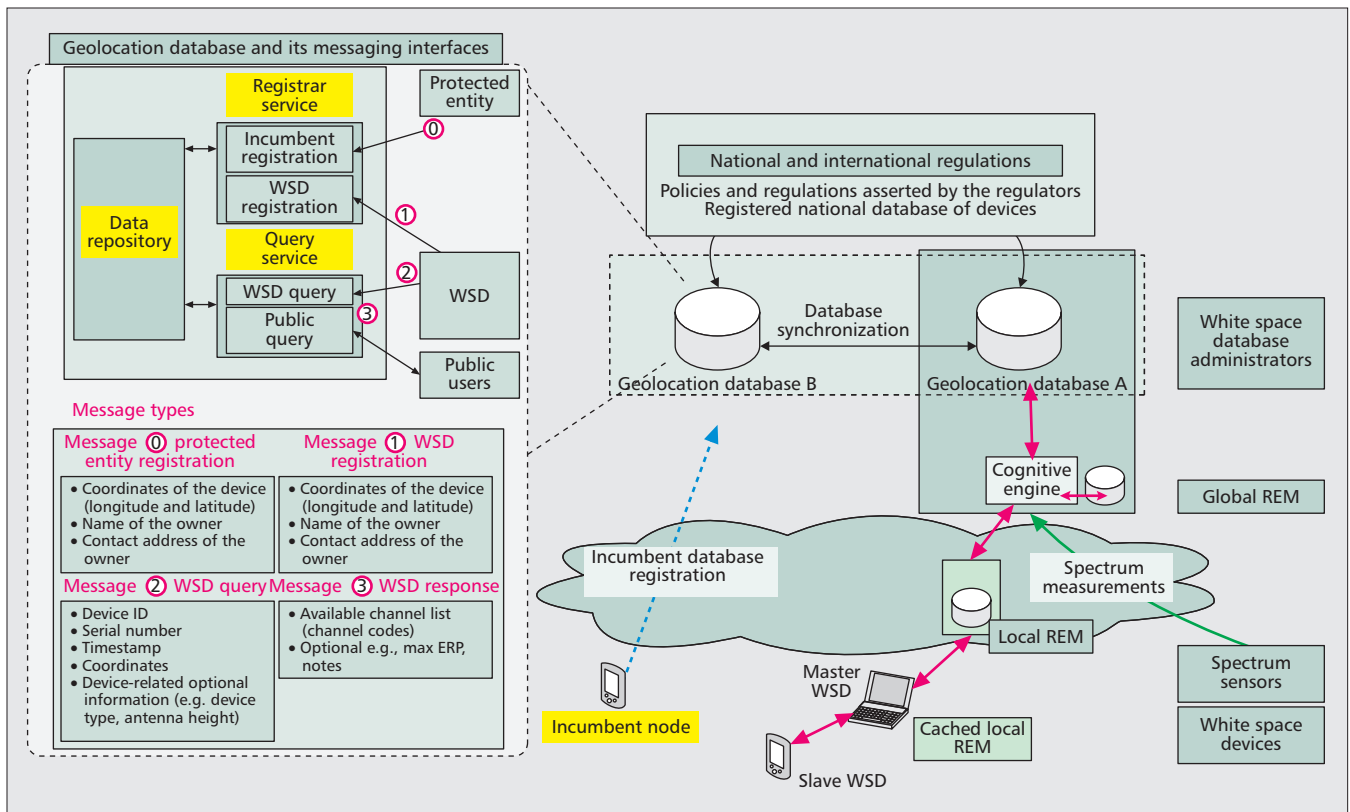
### BASICS OF THE REM CONCEPT

REM was first defined as an abstraction of a real-world environment storing multi-domain information (e.g., PUs, policies, terrain data) [4]. REM maintains information about

- Radio elements
- Environment
- Radio scene [5]

The main difference of REM from the geolocation DB is that REM generates spectrum map by processing the data collected from multiple sources (e.g. measurements from MCDs and regulations from the regulatory bodies in a country) with its cognitive engine, and therefore can easily adapt to dynamic operating environments whereas geolocation DB stores quasi-static information. REM introduces environment awareness that would be harder to acquire by individual CR capabilities via extensive spectrum analysis. Hence, REM can also be seen as the network support turning simple nodes into intelligent

<sup>1</sup> <http://whitespaces.spectrumbridge.com/whitespaces/>,  
<http://www.ukfree.tv/>,  
<http://whitespaces.msresearch.us/> and  
<http://www.tvfool.com>.



**Figure 2.** An overview of REM and its components including the geolocation-database services and message formats adapted from [2].

CRs. Simplifying devices leads to cheaper devices, which in turn eases the commercialization of CRs. Moreover, REM eases the adaptation of a CR device to a new environment (e.g., a new country where regulations are different) in which the individual CR has not constructed any knowledge base yet [5]. Therefore, REM is a promising concept for efficient CRN operation without extensive burden on CRs and it can be linked with the cognitive engine located at the network.

Figure 2 depicts a simplified overview of REM architecture utilizing TVDB and its interfaces of TVDB with WSD. As illustrated in the figure, an incumbent can register through a web interface to one of the geolocation databases. REM, depicted as a module consisting of geolocation database and a cognitive engine in the figure, receives the request from the master device and replies with the optimal operation parameters. The encapsulated cognitive engine can derive the underlying facts on the network and PU/CR traffic via mining the collected data from sensors and using the toolboxes such as statistical toolbox proposed in FARAMIR [6].

Currently, geolocation databases have come into reality by FCC's intent for TVWS access through databases. On the other hand, REM still needs to be explored further: what kind of data to store, how to store, and how to utilize data for cognition are yet to be answered.

#### ARCHITECTURE OF REM IN A CRN

A CRN may deploy both local REMs and a global REM. The global REM provides extensive processing capabilities whereas the local REMs increase the responsiveness of the system [5].

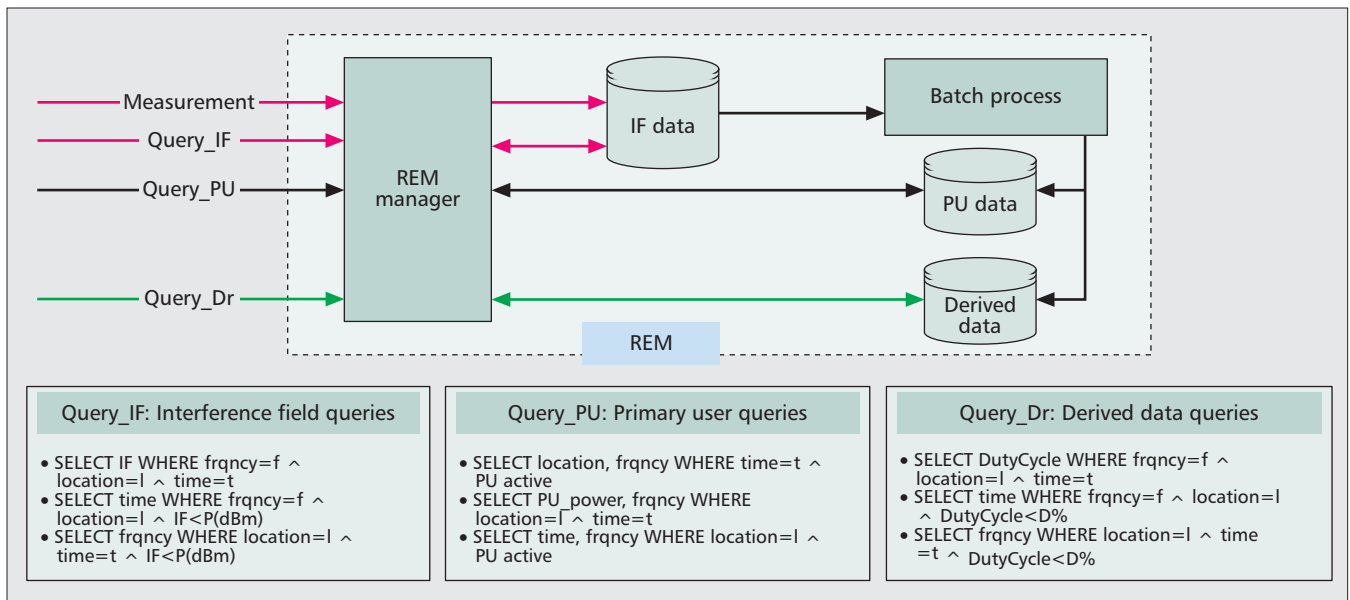
The scale of the global REM depends on the size of network's service area whereas the scale of local REM is determined by the unit service area, e.g., a cell for a cellular network. Therefore, space complexity of REM depends on the number of nodes, number of channels, size of network coverage, granularity of REM information, etc [4]. Local REMs synchronize with the global REM and also update it via exchanging their experiences and observations. Synchronization of REMs is vital for the accuracy and reliability of the information provided. Data must be disseminated efficiently to avoid the information becoming obsolete or outdated. Similarly, since the available spectrum and regulations may depend on the country, each country can have its own map that is in harmony with the neighbors in the border areas. Moreover, each CR node can keep its personal cached REM based on information retrieved from local REM and its experiences.

#### REM DATA MODEL

REM stores information on

- Radio elements
- Environment
- Radio scene [5]

Radio elements contain all the information related to all devices that are capable of transmitting and/or receiving radio signals. Transceiver properties, locations, and their mobility states are examples of this class. Radio environment characteristics contain the interference field data in a specific area plus all the related information such as geographical properties and terrain model. Radio scene data provides a wide range



**Figure 3.** REM data flow and example queries.

of information about the operation of the networks such as provided services, policies as well as operators in an area.

Alternatively, we can classify the REM data into three as *static*, *volatile*, and *derived*. Static data corresponds to the information that does not change frequently, e.g., the locations of TVWS towers for a TVWS network, or cellular operators in a region. On the contrary, volatile data represents the information that is highly dynamic. CRs in a network and their spectrum usage information can be named under this class. Lastly, derived data is interpreted from static and volatile data by some processing techniques. PU on/off activity patterns, and duty cycles can be given as derived data examples.

The design of the REM relies on the following related processes: data gathering and representation, data processing/fusion, and data retrieval/query.

**Data Gathering and Representation** — Efficient representation of information content of a REM is the key to the operation of CRNs. A measurement report by an MCD should contain the following information:

- Received primary (and/or secondary) signal levels
- Location of the measurement
- Time instance
- Inspected frequency bands

MCDs transmit their measurements to the functional entity dubbed as *REM Manager* [5]. In the context of spectrum sensing, nodes can either perform spectrum measurements and report them to the REM Manager periodically, or REM can trigger (some of) the nodes to perform spectrum sensing. This step, just like the spectrum sensing step in cognitive cycle, is the essential step for environmental awareness.

**Data Processing/Fusion** — REM can extract various information about the system via processing the collected volatile data and static

data. In other words, REM manager fuses the data received from MCDs/WSDs and geolocation DB to have a complete understanding of the environment. Figure 3 shows the data flow between the CRs and the REM. REM manager processes the collected data together with the information from geolocation DB to derive PU information and various parameters using various techniques such as radio element parameter modeling, estimation of the transmitter activity patterns, and spatial statistics for radio element modeling.

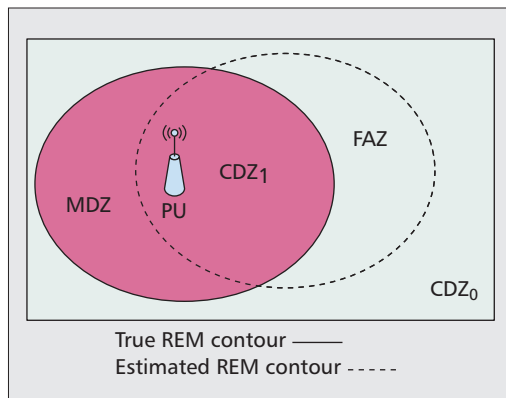
Previous research [7, 8] has shown that CRs can benefit from their knowledge on transmitter activity pattern for channel access. If CR knows or estimates the probabilities of channels' being idle, it can select the channel with the highest probability. Alternatively, if PU activity and inactivity periods are known, CR can select the most appropriate channel accordingly during the spectrum decision phase of the cognitive cycle.

**Data Retrieval/Query** — Before initiating a communication, CRs consult the REM to be informed about the optimal and permitted operation parameters via submitting queries to the REM manager. Since REM maintains both the historical and current spectrum occupancy information, CRs can query to acquire both the current occupancy status as well as some statistical information, e.g., activity duty cycle and raw historical data of the channel parameters. Figure 3 presents some very basic example queries (e.g., about the interference levels or PU duty cycle information) that can be submitted to the REM Manager.

## REM CONSTRUCTION TECHNIQUES

REM must provide reliable information in order to meet the regularity restrictions on non-harmful PU operation. Besides, stored data should be highly accurate and up-to-date (e.g., especially for volatile and derived data) for the maximal

The frequency or volume of measurements, required accuracy in measurements, and density of measurement points are key factors determining the performance of each method. Although Kriging requires more measurement points, it is the most commonly applied technique in the literature due to its higher precision.



**Figure 4.** False alarm, missed detection, and correct detection zones.

network efficiency which calls for a *high quality REM* (which will be detailed later). Hence, REM must be constructed either periodically or upon a request, on a timely manner capturing the network dynamics. REM construction is a broad concept referring to creating a complete map of CRN coverage area. However, we will only focus on deriving interference level at each “pixel” of the CRN. This is also referred to as Interference Cartography (IC).

In the literature, REM construction techniques can be broadly put into two classes: *spatial statistics based methods* and *transmitter location determination based methods*, which are also referred to as *direct* and *indirect* methods, respectively [5]. In the following sections, we overview these methods.

### SPATIAL STATISTICS BASED METHODS

Spatial statistics describe the statistical properties of a given area utilizing the spatial correlational structure of this region. Using spatial statistics and given the measurements at specific locations, missing data at areas without any measurements can be estimated as a function of measured data. Estimation is based on the basic principle that geographically near locations are more related to each other compared to the more distant locations. Possible estimation techniques in the literature are Kriging, Inverse Distance Weighted (IDW), and nearest neighbor interpolation. These methods assume the values of interest at unexplored locations can be estimated using weighted linear combination of the measurements at known data points.

Given that the measured values ( $z_i$ ) at the points  $p_i$ , an interpolation technique basically takes these values as input and estimates the value  $\hat{z}(p)$  at point  $p$ . IDW uses the weights inversely proportional to the distance between  $p$  and  $p_i$  for weighting  $z_i$  values and evaluates the weighted sum of measured values for calculating  $\hat{z}(p)$ . IDW with power  $\beta$  uses  $w_i(p) = d(p, p_i)^{-\beta}$ , where  $d(., .)$  is a distance function and  $w_i(p)$  is the weight of  $z_i$  for evaluating estimate value at point  $p$ . For  $\beta = 1$  and  $\beta = 2$ , the method is referred to as IDW1 and IDW2, respectively.

Kriging, originally used in mining exploration, has been applied in multitude of domains consisting of environmental science, meteorology, agriculture, and remote sensing. Kriging basic-

ly weights the measurement points to find the estimated value for point  $p$ , where weights are the solutions of a system of linear equations that minimize error of prediction. The weights used in linear combination depend on spatial correlation derived from *semivariogram* model of the measurement data. Semivariogram determines the correlation between any two points in the considered system based on their distance separation [9].

The frequency or volume of measurements, required accuracy in measurements, and density of measurement points are key factors determining the performance of each method. Although Kriging requires more measurement points, it is the most commonly applied technique in the literature [10] due to its higher precision.

### TRANSMITTER LOCATION DETERMINATION BASED METHODS

Aforementioned statistical methods directly approximate the signal strength without being concerned about the sources of the power. However, if information on PU locations are already available or can be approximated, they ease the process of REM construction. Transmitter location determination based methods use this approach and first focus on localizing the transmitter(s) and deriving their properties. Subsequently, they estimate the signal strength at each location by applying propagation modeling. However, this approach has more degrees of freedom: multiple transmitters, transmitter properties such as antenna propagation pattern, and accurate characterization of the propagation environment.

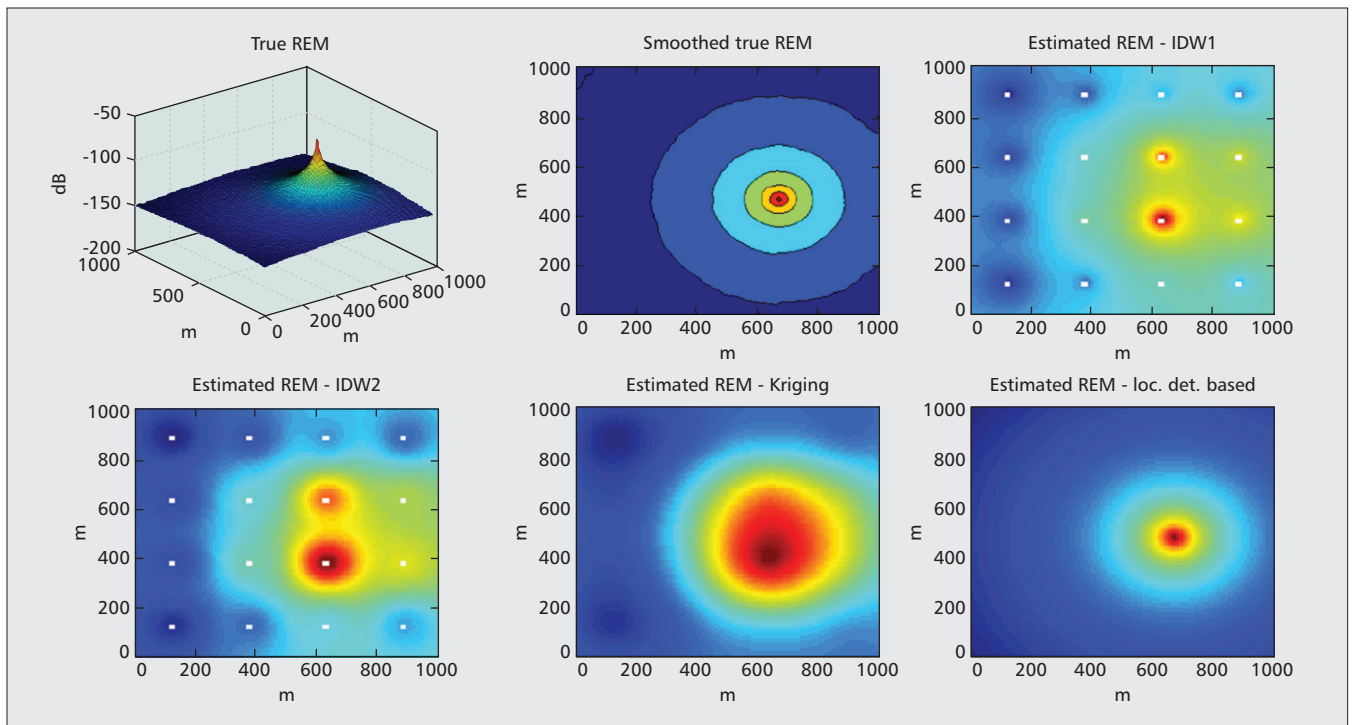
In [11], Riihijärvi *et al.* study the effect of transmitter properties on the signal strength in a CRN using the second order statistics. In [12], a novel transmitter location determination based REM construction technique is proposed and compared with known techniques. Reference [13] applies an image processing based technique, which identifies the transmitters in the system and estimates their parameters based on RSS measurements from sensors.

### REM QUALITY

How accurate a REM describes the real operation environment is measured by REM quality metrics. These quality metrics can be evaluated through comparing the *estimated REM* with the *true REM*. Number of sensors,<sup>2</sup> their distribution in the network and sensing accuracies, dynamics of the propagation environment, and accuracy of the propagation modeling are the key parameters affecting the performance of the REM construction methods.

One of the ways used in the literature to quantify the similarity between the true and the estimated REM is average Root Mean Squared Error (RMSE) calculated at each grid point as the difference in RSS between the estimated REM and the true REM [14]. However, this metric falls short of identifying in which regions the spectrum opportunities are lost because of over-estimating the RSS and thereby forbidding the CR communications as well as in which regions the interference levels are under-esti-

<sup>2</sup> We use “sensor” as a general term for MCDs and CRs which can perform spectrum sensing.



**Figure 5.** Constructed REMs (path-loss exponent = 3.5; path-loss correction = 38.4 dB; shadowing  $\sigma_s = 8$  dB).

mated and thereby leading to CR collisions with on-going transmissions in the region. Hence, Correct Detection Zone Ratio and False Alarm Zone Ratio are introduced in [12].

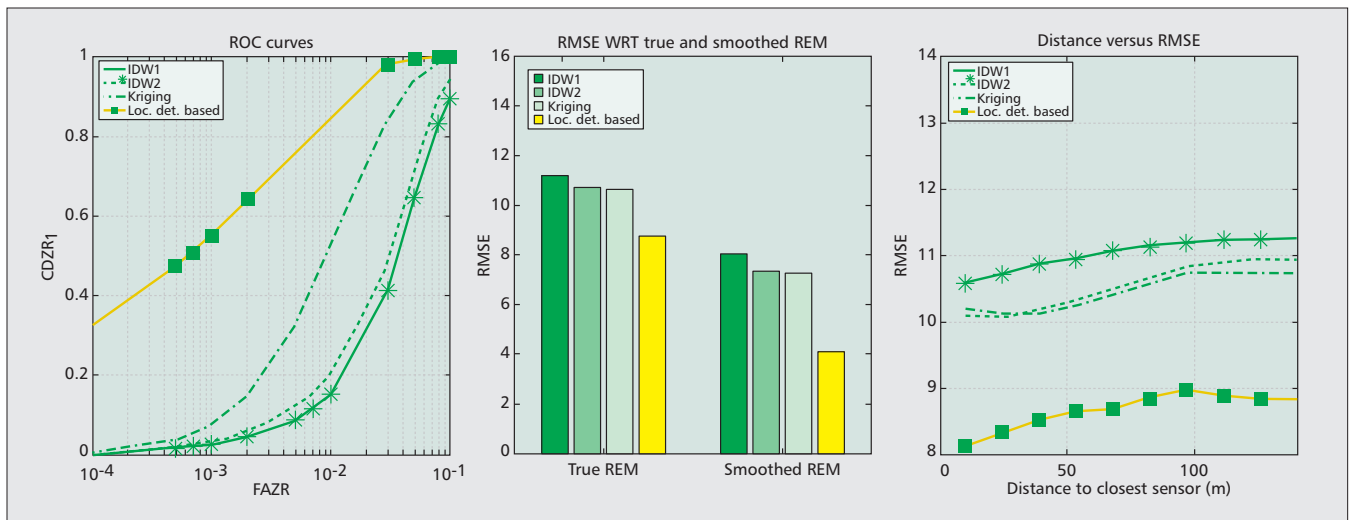
The aforementioned regions in which opportunities are lost or collisions occur can be determined by analyzing the intersection zones of the true REM and the estimated REM. Figure 4 depicts the true REM and the estimated REM contours. The solid circle divides the area into two subareas. Inside the true REM contour (red region), secondary transmission is forbidden. The areas where the true REM and the estimated REM contours both determine forbidden or allowed zones correctly are called the Correct Detection Zone type-1 ( $CDZ_1$ ) and Correct Detection Zone type-0 ( $CDZ_0$ ), respectively. False Alarm Zone (FAZ) is the area in which transmission is forbidden due to estimation errors but the area is actually the allowed zone for transmission. Finally, Missed Detection Zone (MDZ) is the area where transmission is not allowed but estimated REM infers the opposite. We normalize the regions according to true REM zones, and determine the  $CDZ_1$ , FAZ, and MDZ ratios ( $CDZR_1$ , FAZR, and MDZR), which present a better understanding about the performance of a REM construction method. FAZR is defined as the ratio of the area denoted by FAZ to the total area covered by FAZ and  $CDZ_0$  while  $CDZR_1$  is the area of  $CDZ_1$  normalized by total area covered by  $CDZ_1$  and MDZ.  $CDZR_0$  and FAZR are analogous to probability of detection and false alarm in the spectrum sensing context, respectively.

In a simple scenario, we analyze the performance of various techniques. We consider an area of 1000m  $\times$  1000m divided into subregions with an edge size of 250m and a sensor in the

center of the subregion. There is a single PU located at (670m,470m), and the channel between each sensor and the PU has log-normal shadowing. Each sensor measures the signal strength and reports it to the REM Manager. Figure 5 illustrates the true REM and the constructed maps. Smoothed true REM stands for the bird's-eye view of the filtered true REM. IDW based techniques have diverging estimations (infinity since  $d = 0$ ) at sensor locations resulting in white points in the figure. We use IDW1 and IDW2 for inverse distance and distance square weighted interpolation methods, respectively, and Loc.Det for transmitter location determination based REM construction.

As ROC curves in Fig. 6 show, for a given FAZR the corresponding  $CDZR_1$  is significantly higher for Loc.Det. since it has extra information (i.e., channel parameters) compared to other methods. IDWs perform similarly and with noticeably lower performance compared to Kriging. RMSE figure in Fig. 6 supports our previous claim that RMSE may not suffice for performance analysis of the REM techniques. From the figure, we see that the performance difference between Kriging and IDW2 are not apparent, however it can be seen clearly in ROC curves (Kriging performs significantly better than IDW2). RMSE aims to represent the divergence from the true REM without being concerned about the type of the error, i.e., false or missed detection.

We also analyze these methods according to their sensitivity against the minimum distance between the PU and a sensor. As overviewed earlier, some techniques (e.g., IDW based and Kriging) heavily depend on the PU distance to the closest sensor (refer to  $d(p, p_i)$  in the calculations). Hence, if a sensor is very close to the



**Figure 6.** Different metrics for performance evaluation (path-loss exponent = 3.5; path-loss correction = 38.4 dB; shadowing  $\sigma_s$  = 8dB).

PU, then its measurement will dominate the estimation resulting in a very accurate estimate. For such techniques, it is paramount to deploy enough many sensors in a uniform manner to achieve a reasonable performance. Figure 6 shows the robustness against PU closeness to the sensors. Deviation from the true REM increases with the increase in the minimum distance. Again, IDW2 and Kriging perform very similar according to the figure, however in ROC curve analysis Kriging performs significantly better than IDW2. Hence, different metrics must be considered for evaluating REM construction techniques. Performance ordering of the methods in terms of metrics may be similar, however the quantitative and relative performance differences may change.

In our analysis, we model a simplistic case which ignores mobility of the PU as well as the terrain model in order to focus on the REM quality metrics without getting lost in the complexities of the scenario. Depending on the REM update interval and the PU's mobility, change in the location of the PU may be negligible or not. For example, if the PU is a pedestrian, then previously constructed REM will be more similar to the true REM of a few seconds later, compared to the case where the PU is in a car. Deciding on the REM update intervals is one of the challenges listed in the next section.

## RESEARCH CHALLENGES

REMs have strong bonds with cognitive engines in the network and have the potential to realize real intelligent networks. However, research on REM is yet to be improved to tackle the following challenges:

**Algorithmic complexity:** As a REM construction technique must be performed online upon a request from a CR, it must be computationally easy and should respond timely. However, these requirements also bring a tradeoff regarding accuracy. If REM construction is performed less frequently, depending on the dynamics of the network, REM information may become obso-

lete and acting based on this outdated information may result in inferior operation. Therefore, tradeoff between accuracy and efficiency must be balanced.

**Scale of REMs:** As presented in Fig. 2 REMs can be deployed at different localities: local REM for a small-area coverage and global REM for a wide (possibly country-wide) coverage. However, there is no obvious answer for the optimal scale of these local REMs. It is yet to be researched how to deploy these entities for maintaining a reliable REM while avoiding too many deployments due to the cost and the algorithmic complexity that is proportional to the size of the REM coverage area.

**Indoor geolocation:** REM techniques rely on geolocation information. However, indoor localization is a challenging issue due to incapability of satellite-based positioning which is known to be a highly accurate method.

**Propagation modeling:** Performance of REM construction techniques is directly affected by the accuracy of propagation modeling.

**REM synchronization:** REMs must present current information. Hence, any information in one of the REMs (e.g., global or local) must be synchronized with the other REMs. However, information dissemination should strike a balance between information freshness and overhead of dissemination.

**Interoperability and security:** REMs must be secure, spoofing-free and must meet reliability requirements.

**Standardization:** Standards for country-agnostic or spectrum-agnostic access to geolocation databases and REMs need to be developed.

## CONCLUSIONS

REM, being a key component for efficient realization of TVWS communications and enabling cognition in these systems, is the main focus of this article. REMs are envisioned as the crucial building blocks of CRNs since they provide various information about a network including the basic information on the physical environment,

regulation, policies, typical PU activity patterns, and PU profiles. Therefore, REM can be seen as a fundamental part of the CRN cognition and facilitates the coherent coexistence of PUs and CRs.

In this work, we present different aspects of REM entailing the REM architecture, data stored in REMs, REM construction methods, and their performance measures. Performance measures of the REMs tell different properties of the construction techniques. In addition, we compare performance of various REM construction algorithms in a basic illustrative scenario. As an efficient and feasible enabler for opportunistic spectrum access, REM presents an opportunity for paving the way envisaged CR systems to practical networks. However, research on REMs is still at its early stages and there are many intertwined challenges to be achieved such as algorithmic complexity, indoor geolocation, and REM synchronization.

#### ACKNOWLEDGMENT

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